To
Date: 28/01/2025

To,
Dr Palas Kumar Farsoiya
Department of Chemical Engineering
Indian Institute of Technology, Roorkee

Email Id: palas.farsoiya@ch.iitr.ac.in

Dear Sir,

Subject: Budgetary Proposal for Stratasys J35Pro Polyjet 3D Printer

With reference to your ongoing discussions with me **(Mr. Sanjeev Kumar)**, attached is the techno-commercial budgetary proposal for **Stratasys J35 Pro Polyjet 3D Printer**.

The proposal is complete with technical literature indicating system specifications, process workflow and reference case studies of Indian and global customers using these machines for prototyping, tooling, and production applications, along with commercial offer indicating system, accessories, training, and consumables.

We hope that same is in line with your expectations and should you have any further queries, feel free to get in touch with us.

We look forward to a favourable response.

For
Adroitec Information Systems Pvt Ltd.,

Sr Manger- Sales

Sanjeev Kumar
+91-8000600042
Sanjeev.kumar@4dsindia.com

INDEX OF ENCLOSURE

S.NO	DESCRIPTION	ANNEXURE
1	About 4DS Adroitec & Stratasys	A
2	PolyJet Modelling – Technology Brief	B
3	J35 Pro – Materials and Capability	C
4	Polyjet Technologies – Applications and case Studies	D
5	System Specification – J35 Pro Polyjet Modelling System	E
6	Investment Details - Stratasys Polyjet J35 Pro	F
7	Commercial Terms of Business	G

ANNEXURE-A

About 4D Simulations

4D Simulations, a unit of Adroitec Information Systems Pvt. Ltd., is dedicated to the world of Additive Manufacturing. We are exploring new aspects of 3D printing on daily basis and providing 3D printing solutions for various industries and areas of interest.

We as a partner for various international principals, innovate, develop and work as a better solution provider in the market for most of the 3D printing technologies like Fused Deposition Modelling, Stereo Lithographic Addition, Direct Metal Laser Sintering, Direct Energy Deposition, and so on.

We also provide 3D printing & additive manufacturing services to the automotive, defense, Aerospace, Drone & UAV, Healthcare, Medical device companies, and consumer goods industries. We also doing research on different materials as per customer application requirement.

We, a 34+ years old group, continuously working on every aspect of design and development and have experience from old traditional engineering and manufacturing knowledge to recent additive manufacturing and smart engineering practices. We welcome our customers to work side by side and grow with us and discover the flavors of this new technology of Additive Manufacturing. Our Expertise ranges in CAD, Reverse Engineering, Concept Designing, 3D Printing, Tooling, Digital Creation, Post Processing of 3D Printed Parts, Rapid Prototyping, and so on.
We as a brand provide our customers, with the technology that suits best for their research and development and prototyping & for their personalized and engineering purposes and also help them in innovating and discovering new aspects of their business in the field of additive manufacturing. This helps our customers get more productivity in lesser time, more prototyping sessions in short intervals, equip them with recent technologies, smart reverse engineering methodologies, and great business growth.

Our Centre is PLI qualified under the government of India notification No. CG- DL-E-30/09/2021- 230076
SN.NO. 4044 (E-date 30/09/2021)

About Stratasys:

Stratasys is leading the global shift to additive manufacturing with innovative 3D printing solutions for industries such as aerospace, automotive, consumer products and healthcare. We are re imagining what's possible in 3D printing, reaching for new levels of realism, accuracy, speed and performance. Through smart and connected3D printers, polymer materials, a software ecosystem, and parts on demand, Stratasys solutions deliver competitive advantages at every stage in the product value chain. The world's leading organizations turn to Stratasys to transform productdesign, bring agility to manufacturing and supply chains, and improve patient care.

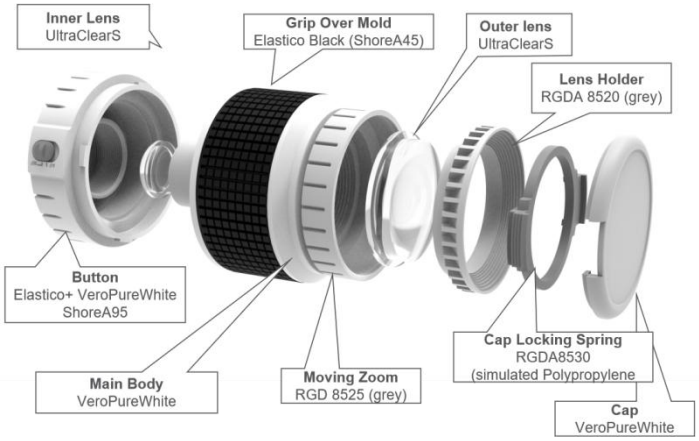
ANNEXURE-B

Polyjet Modelling – Technology Brief

Invention of the Polyjet technology

The Polyjet technique was developed by the company Objet Geometries Ltd. Objet was founded in 1998 by **Rami Bonen**, **Gershon Miller** and **Hanan Gotaiit**.

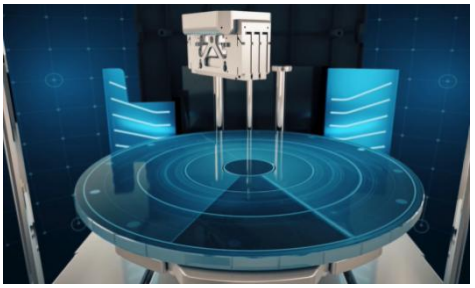
In 2008, the company presented the first multi-material 3D printer. In 2012, the first 3D printer with more than 100 combined materials was introduced by Objet Ltd. **In the same year, Objet merged with Stratasys Inc of USA forming Stratasys Ltd.**



Physics of Polyjet

Photo polymeric Plastic/rubber like material is jet through a series of nozzles and cured using UV Light.

Polyjet J35 Pro machine can mix up to 3 materials at once and create nearby 10,000 greyscale colors and material combinations in the same part.

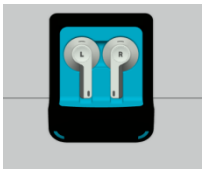


Polyjet is an office friendly 3D Printing process which works similar to an inkjet 2D Printer, where instead of jetting ink, the printheads jet tiny droplets of Photopolymer resin (Opaque, Transparent etc) with a layer thickness of 18 microns which is cured instantaneously using UV lamps placed along the side of the Printhead. The overhangs are supported using Non-Toxic Gel based Support Material which can be cleaned using Waterjet. Parts once printed and cleaned are ready for use with no post curing required.

.

Polyjet Workflow

Design



Material Assignment – Print Preparation

Printing

Support Removal with Medium Pressure Waterjet

Final Part (Ready to Use)

ANNEXURE-C

Polyjet Technology - Photopolymer Materials:

Rigid and Opaque Materials:

PolyJet rigid opaque (Vero Family) materials in **montone colours give brilliant colour** finish and includes:

VeroUltraWhite™
VeroUltraBlack™

Ideal For:

- Fit and form testing.
- Moving parts and assembled parts.
- Sales, marketing, and exhibition models
- Assembly of electronic components.
- Silicon moulding Patterns

Rigid and Transparent Material (Both):

Vero Ultra Clear lets you prototype **clear parts**, the material is designed to mimic Glass/PMMA with **transparency of 90%.**

Vero Ultra Clear **can be mixed with Vero Colours to give tints and can be over-moulded with Polyjet Elastomers.** This can be used to visualize **internal geometries.**

Ideal for:

- Form and fit testing of see-through parts like auto lighting consumer products, eyewear, light covers and cases.
- Visualization of liquid flow.
- Visualization of Internal Cross Sections.
- Medical Models.

Engineering Materials

Digital ABS offers toughness combined with exceptional dimensional stability enabling users to combine realism and aesthetics with functional characteristics and can be over moulded with elastomers.

It suitable for high stress and thermal applications such as moulding and forming.

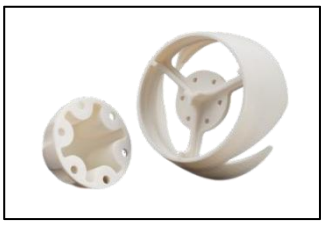
Ideal for:

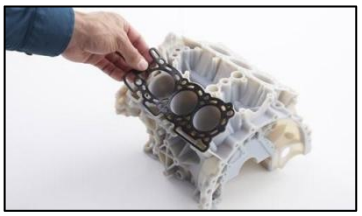
- Functional prototypes
- Snap-fit parts for high or low temperature usage.
- Electrical parts and casings

Elastico ShoreA50 (Polyjet Elastomer) rubber-like materials are designed to mimic elastomeric properties. With a base shore hardness of 50A, it can be mixed with Vero Family/Digital ABS to form parts with variable shore hardness, flexibility, and mechanical properties. and create Flexible Plastics (to mimic properties of Polypropylene).

Ideal for:

- Rubber surrounds and over-moulding
- Knobs, grips, pulls, handles, gaskets, seals, hoses





ANNEXURE-D

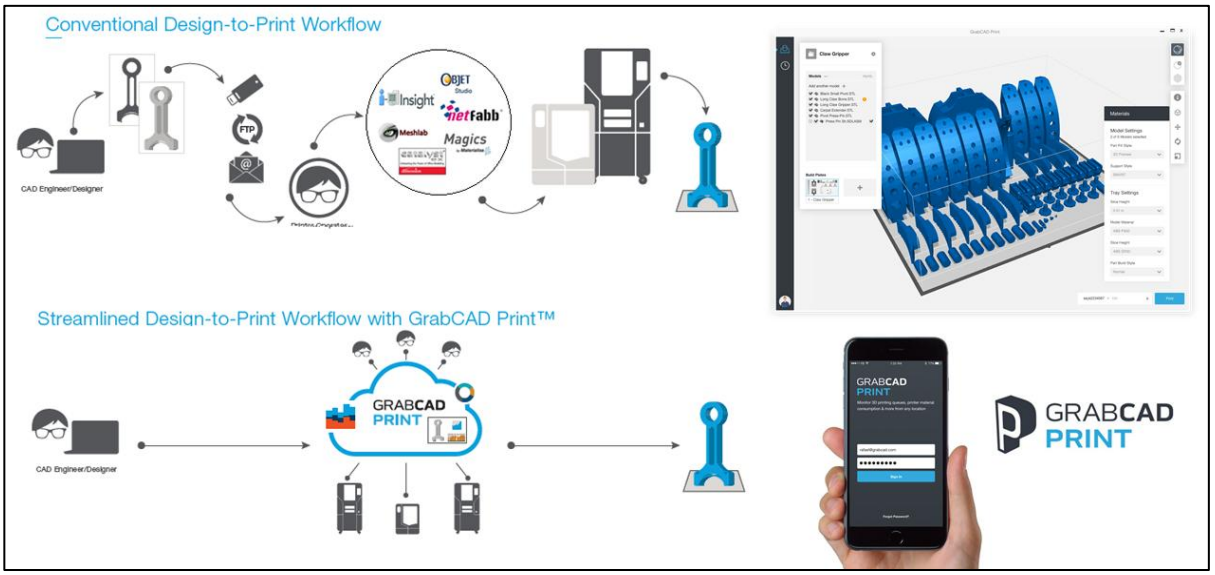
Polyjet Technology – Applications

When 3D Printing was invented in 1989, It was developed to help designers perform basic prototyping tasks such as form and fit studies, however with the improvement in machines, materials, design practices and post processing methods. 3D Printing is moving from Prototyping to Tooling and to production and hence finding its place in all stages of a product life cycle.

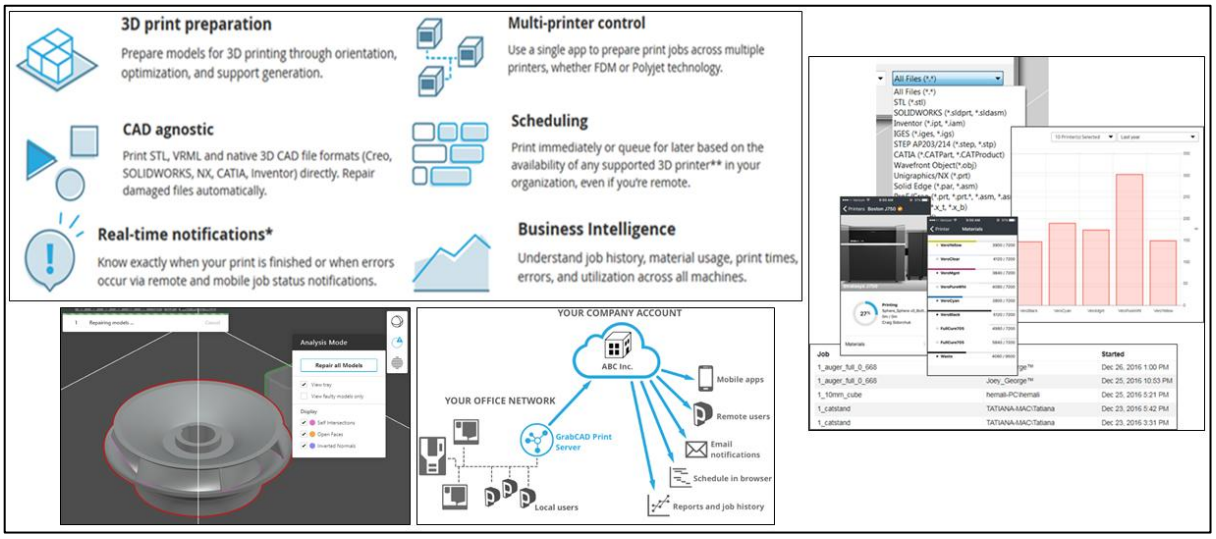
Effective and Efficient Workflow with GrabCAD Print.

GrabCAD Workflow

The most powerful Enterprise 3D Printer becomes more powerful coupled with the most advanced 3D printing software: GrabCAD print allows users to work directly with Native CAD files, with the ability there is no need to use 3rd party software to convert Native CAD files to STL's and repair them. GrabCAD print can import most native CAD formats and perform repair operations to fix issues such as Inverted Normals, Open surfaces etc.



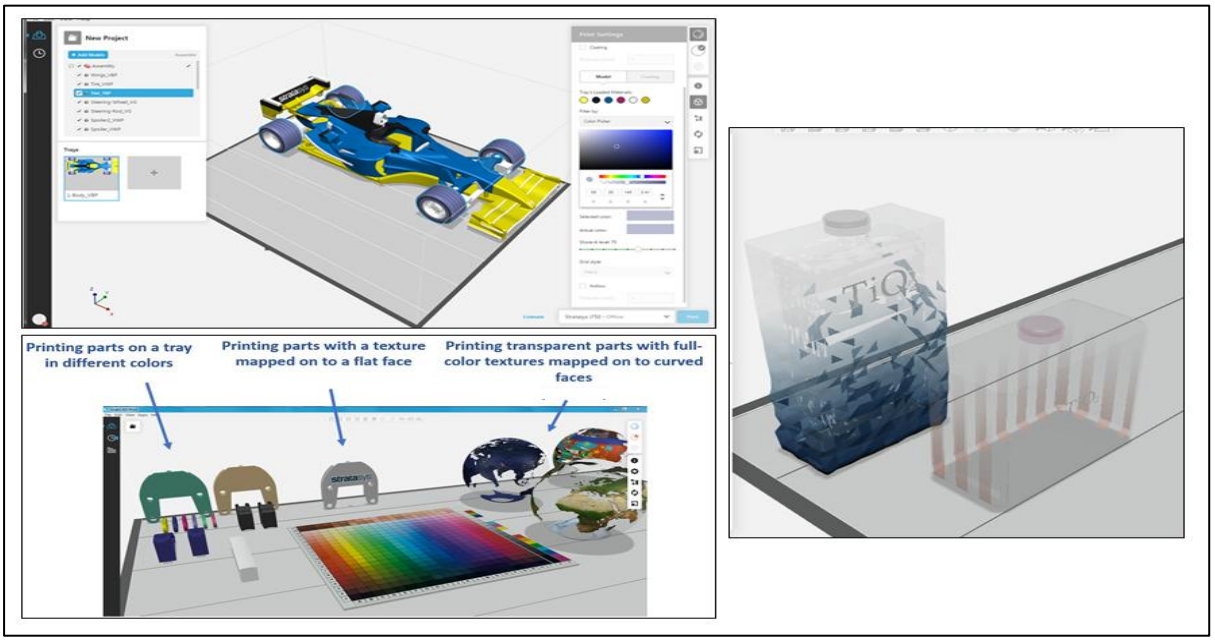
With GrabCAD Print, Users can manage their printer/printers and generate reports, manage jobs and monitor/control the printer via the internet/network.



CMF Assignment with GrabCAD

GrabCAD Print allows user to assign colours and materials in 3 different ways:

- Impost Assembly CAD and assign Material/Colour by Colour-Material Selector.
- Import VRML/OBJ files with preassigned finish, transparency and/or colour.
- Import Part/Assembly and Assign Colour from Pantone Library



ANNEXURE-E

System Specification – J35 Pro Polyjet Modelling System:

Technology:	Polyjet	
Build size:	X-Y Area (Circular Bed)	1,174 cm2
	Z Height	155 mm
Layer thickness:	Horizontal build layers down to 18.75 microns (High Quality Speed Mode)	
Accuracy:	Deviation from STL dimensions, for 1 Sigma (67%) of models printed with rigid materials, based on size: under 100 mm – ±150µ; above 100 mm – ±0.15% of part length. Deviation from STL dimensions, for 2 Sigma (95%) of models printed with rigid materials, based on size: under 100 mm – ±180µ; above 100 mm – ±0.2% of part length.	
Size and weight: Machine:	651 x 661 x 774mm | 98 kg	
Workstation compatibility:	Windows 10.	
Network connectivity:	Ethernet TCP/IP 10/100 base T	
Power requirements:	100-240VAC, 50-60 HZ, 10A, 1 phase	
Regulatory compliance:	CE, FCC, EAC, cTUVus, CB, RCM	
Operational environment:	Temperature 18 C-25 C (64 F-77 F); Relative Humidity 30-70 % (non-condensing)	
Model Materials:	**Rigid and Opaque Materials (Vero Ultra):** BlackS and WhiteS. **Rigid and Transparent Materials:** Vero Ultra ClearS (Transparency: 90%) **Draft Grey:** Print at Twice the speed. **Digital Materials:** 3 Materials can be mixed simultaneously to create **multiple combinations and greyscale parts.** ***Engineering Materials :*** *Elastico ShoreA50 (Elastomer; Clear & Black) and Digital ABS (High Temperature)*	
Support Material:	Non-Toxic Gel like support material (SUP710) -Removable with Waterjet and WSS150 (Water Soluble Support Material).	
Software:	GrabCAD Print.	
Noise Specifications:	46 dB maximum during build, 35db when idle	

ANNEXURE-F

Stratasys J35 Pro 3D Printer

Product Code	Product Name	Description	QTY	Total Price
OBJ-19002-V	J35 PRO PRINTER PACKAGE WITH PROAERO/ PROBLEACHER	J35 PRO PRINTER PACKAGE WITH PROAERO/ PROBLEACHER	1.00	Rs 63,53,718
OBJ-19000	STRATASYS J35 PRO 3D PRINTER	STRATASYS J35 PRO 3D PRINTER	1.00	
OBJ-09131	PACK OF 1 RGD821, VEROULTRACLEARS, 1.1KG	PACK OF 1 RGD821, VEROULTRACLEARS, 1.1KG	1.00	
OBJ-09125	PACK OF 1 SUP710, SUPPORT, 1.1 KG	PACK OF 1 SUP706- L, SUPPORT, 1.1 KG	1.00	
OBJ-09120	PACK OF 1 CLEANSER, CLEANING FLUID, 1.1KG	PACK OF 1 CLEANSER, CLEANING FLUID, 1.1KG	1.00	
OBJ-09133	PACK OF 1 RGD824, VEROULTRAWHITES, 1.1KG	PACK OF 1 RGD824, VEROULTRAWHITES, 1.1KG	1.00	
OBJ-09132	PACK OF 1 RGD864, VEROULTRABLACKS, 1.1 KG	PACK OF 1 RGD864, VEROULTRABLACKS, 1.1 KG	1.00	
OBJ-09007	J55/J35 HOW SCALE KIT	Head optimization wizard (HOW) scale kit for J55™/J35™ 3D printers. The kit allowing an efficient Print Head drop weight calibration.	1.00	
OBJ-09400	PROAERO AIR EXTRACTOR, EU/JP/INT	The PROAERO™ incorporates an externally mounted plenum system which effectively captures fumes generated during the printing process and returns ultraclean, filtered air to the operator's breathing zone. It has the benefits of low noise, low power consumption and integral speed control. An optional item for the J55™ Printer system. Base Product – 16.3 x 10.2 x 61 Inch (413 x 260 x 297mm) Envelope. Gas filter – Treated Active Carbon. Filter Particles – 99.997% @ 0.3microns Includes: 1 year warranty.	1.00	
OBJ-09006	J55/J35 SCANNER KIT	Head calibration scanner kit for J55™/ J35™ 3D printers. The kit allowing an efficient Print Head alignment calibration.	1.00	
OBJ-19002	J35 FUNCTIONAL LICENSE	J35 FUNCTIONAL LICENSE	1.00	
GC-SLPROSHIELD-V	GC-SLPRO WITH SHIELD KIT	GC-SLPRO WITH SHIELD KIT	1.0	
GC-SLPRO	STREAMLINE PRO Duration: 1Year	STREAMLINE PRO 1 printer license	1.0	

OBJ-05500	KIT, STREAMLINE PRO SHIELD	KIT, STREAMLINE PRO SHIELD	1.0	
OBJ-19400	PJ PHOTOBLEACHING UNIT 230 V	Use Pro Bleacher - OBJ-19400 PJ PHOTOBLEACHING UNIT 230 V for countries with grid voltage of 210V-250V	1.00	
4DS-01012025	WATER JET SYSTEM, 220V-240V 50HZ	WATER JET SYSTEM, 220V-240V 50HZ	1.00	
			Total Amount	Rs 63,53,718
			Special Discounted Price for **Academic Association with IIT Roorkee**	**Rs 47,65,288**
			GST @18 %	**Rs 8,57,751**
			Total Amount With GST	**Rs 56,23,039**
In Words – Fifty-Six Lacs Twenty-Three Thousand Thirty-Nine Only				

ANNEXURE-G

Commercial Terms of Business

1	*Pricing Currency*	→	Prices quoted INR.
2	*Custom Duty*	→	**Included in above Quote**
3	*GST*	→	Extra as indicated @ 18%, Exemption certificate if any should be enclosed with the Purchase Order.
4	*Freight Forwarding & Insurance*	→	**Included in above Quote**
5	*Proposal Validity*	→	40 Days from the date of the Proposal.
6	*Ordering Details*	→	**Adroitec Information systems Pvt Ltd** **D – 68 Sector 2 Noida 201301** **Adroitec Information Systems Pvt. Ltd.** **A/c No. 38025444368** **GST: 09AAACH2422A1ZB** **STATE BANK OF INDIA** **PLOT NO. - 419, UDYOG SADAN PATPARGANJ INDUSTRIAL AREA, DELHI-110092** **IFSC Code: SBIN0010553**
7	*Fax No: /email*	→	Sanjeev Kumar +91-8000600042 Sanjeev.kumar@4dsindia.com
8	*Delivery*	→	Door Delivery, 3-4 Weeks from Commercially & Technically clear Purchase Order Receipt of Advance Amount, Acceptance of Letter of Credit, whichever is later.
9	*Payment Terms*	→	As per Purchase Order.
10	*Software Licencing*	→	Pre-Processing Software will be perpetual in nature.
11	*Site Requirement*	→	On placement of the Purchase Order, unless otherwise mentioned by Customer, the site for machine is ready for installation as per the "Site Requirement" document attached.
12	*Installation*	→	Installation will be done within one to two weeks from the time of arrival of machine at your premises. Installation is valid for India only on FOC basis by Adroitec and Stratasys
13	*Training*	→	2-3 days of familiarization training on System and Software will be conducted at customer's premises.
14	*Language*	→	English Only.
15	*Warranty*	→	12 Months Comprehensive warranty (Not applicable on Printheads/Consumables/Accessories/Optional Items). The warranty commences from the date of installation and is applicable for parts and software. Customer will get support upgrades during warranty & any parts if failed will be shipped and replaced at no additional cost. Custom duties if any on the replaced part will be borne by Customer. Service Consumables like Print Heads are not covered in warranty.
16	*Machine Specifications*	→	Technical Specifications of the Machine/Systems are provided under relevant heade only. Additionally, the proposal, for better understanding of technology, contains several technical illustrations/ consumable material options etc. which a part of supply may not be.
17	*Confidentiality*	→	Unless indicated otherwise, the information contained in this proposal is privileged and confidential and is intended only for the use of the addressee(s) named above and others who have been specifically authorized to receive it. By receiving this proposal, the recipient acknowledges that he will not engage in unauthorized review use, disclosure, dissemination, distribution, forwarding, copying of this proposal or any action leading to the leakage of the information contained in this proposal which is strictly prohibited and is unlawful.